

# Complementary Material

## 1 Average Correct Answers

Table 1: Pairwise p-values from Dunn’s post-hoc test (Bonferroni corrected) comparing average correct answers per model. The overall Kruskal-Wallis test was significant ( $H = 8.27$ ,  $p = 0.016$ ).

|            | Human  | PhinetunEd | Turbo  |
|------------|--------|------------|--------|
| Human      | —      | 1          | 0.037* |
| PhinetunEd | 1      | —          | 0.040* |
| Turbo      | 0.037* | 0.040*     | —      |

Table 2: Pairwise p-values from Dunn’s post-hoc test (Bonferroni corrected) comparing average correct answers per group and per model

|                                      | PhinetunEd vs Human | PhinetunEd vs Turbo | Turbo vs Human |
|--------------------------------------|---------------------|---------------------|----------------|
| <b>4-1</b> ( $H=12.65$ , $p=0.002$ ) | 1                   | 0***                | 0.001***       |
| <b>4-2</b> ( $H=20.38$ , $p=0.000$ ) | 0.032*              | 0.981               | 0.812          |
| <b>4-3</b> ( $H=11.51$ , $p=0.003$ ) | 0.072               | 0.003**             | 0.857          |

Table 3: Pairwise p-values from Dunn’s post-hoc test (Bonferroni corrected) comparing average correct answers per day. The overall Kruskal-Wallis test was significant ( $H = 10.06$ ,  $p = 0.007$ ).

|       | Day 1 (Foodchain) | Day 2 (Env. Changes) | Day 3 (Collaboration) |
|-------|-------------------|----------------------|-----------------------|
| Day 1 | —                 | 0.15                 | 0.715                 |
| Day 2 | 0.15              | —                    | 0.005**               |
| Day 3 | 0.715             | 0.005**              | —                     |

## 2 Self-reported Enjoyment Scores

Table 4: Pairwise p-values from Dunn’s post-hoc test (Bonferroni corrected) comparing average daily learning scores per group and per model

|                                      | PhinetunEd vs Human | PhinetunEd vs Turbo | Turbo vs Human |
|--------------------------------------|---------------------|---------------------|----------------|
| <b>4-1</b> ( $H=13.97$ , $p=0.001$ ) | 0.028*              | 0.905               | 0.001***       |
| <b>4-2</b> ( $H=1.24$ , $p=0.539$ )  | 1                   | 0.831               | 1              |
| <b>4-3</b> ( $H=2.92$ , $p=0.232$ )  | 1                   | 0.269               | 0.919          |

### 3 Self-reported Learning Scores

Table 5: Pairwise p-values from Dunn’s post-hoc test (Bonferroni corrected) comparing perceived learning scores per group and per model

|                              | <b>PhinetunEd vs Human</b> | <b>PhinetunEd vs Turbo</b> | <b>Turbo vs Human</b> |
|------------------------------|----------------------------|----------------------------|-----------------------|
| <b>4-1</b> (H=2.07, p=0.355) | 0.843                      | 0.517                      | 1                     |
| <b>4-2</b> (H=2.21, p=0.331) | 1                          | 0.446                      | 0.916                 |
| <b>4-3</b> (H=1.72, p=0.422) | 0.828                      | 0.715                      | 1                     |